

Bosch dishwashers: models and reviews analysis

Dario Savaresi,

AIDA for Business Master, University of Milan – Bicocca, July 2026



- Motivations and scenario;
- ETL phase;
- Data analysis;
- Conclusions and next steps.



- **Motivations and scenario;**
- ETL phase;
- Data analysis;
- Conclusions and next steps.

Motivations



R&D



- From a technical standpoint, BSH group represents the state of the art concerning dishwashers, with its European brands: Bosch, Siemens, and Gaggenau.
- R&Ds from all competitors study Bosch dishwashers: technology, cycles, ergonomics, ...
- Quality of Bosch appliances is recognized by the most important consumer magazines in Europe:



Marketing & Product Line



- BSH marketing strategy and customers' perception, instead, is harder to be studied.
- Commissioning studies from external companies is costly, limited in scope, and introduces an unavoidable delay between the date the study is commissioned and the actual delivery of the results.
- Info retrievable from the website is a source of information not adequately exploited.

Scenario & Questions



Scenario



- Bosch exposes its dishwasher catalogue with multi-language reviews on its Italian website.
- Within each dishwasher page, associated claims are shown.
- Characteristics associated to each appliance can be retrieved from HTML pages. Reviews are downloaded thanks to GraphQL APIs.
- The goal is to develop a queryable Data Warehouse containing products and customer reviews.

Business Questions



- What does the typical Bosch customer look like?
- How customer satisfaction varies across countries?
- Which are the most sold products? Which are the products with highest reviews?
- Do claims have an influence on reviews and star rating?

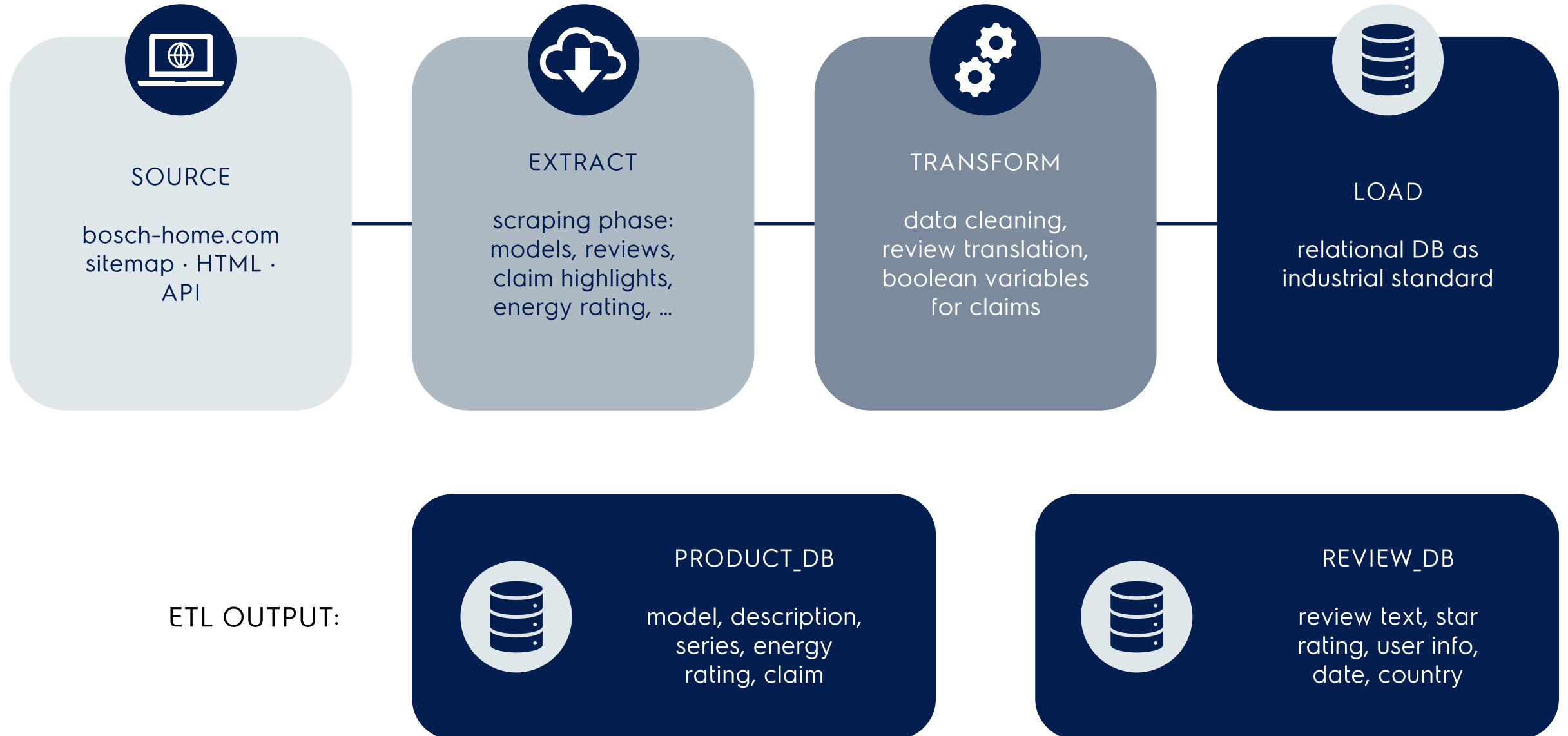
Home Connect: controlla la tua lavastoviglie ovunque ti trovi e sfruttane al massimo tutte le potenzialità.

L'app Home Connect ti permette di controllare la tua lavastoviglie direttamente dallo smartphone, offrendoti la libertà di scegliere il programma ideale senza pensarci due volte. Non sei sicuro che sia in funzione? Ricevi aggiornamenti in tempo reale, per poterti rilassare e concentrarti su ciò che conta davvero. Vuoi ridurre i consumi? Usa la funzione Smart Start e molte altre opzioni intelligenti. È come avere un piccolo assistente che comprende le tue esigenze, rendendo il lavaggio delle stoviglie non solo più semplice, ma anche più smart.



- Motivations and scenario;
- **ETL phase;**
- Data analysis;
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ETL architecture





EXTRACT phase – Scraping

Models

1

Sitemap scanning

Looking for all products of type *dishwasher* with regex: model and URL associated to each product.

2

Dishwasher model scraping

Extraction with BeautifulSoup of base information: name, energy rating, installation type, ...

3

Claim scraping

JavaScript string decoding generated during webpage loading. Concatenation of all claims in a string *claim_raw*.



products.csv

Reviews

1

GraphQL API

For each model, the API call gives a JSON back with all the reviews, not only for the selected model, but also for all the european equivalent models (this is a standard for all manufacturers).

2

Information retrieval

Dictionary-like parsing of information: title, text, user info, rating, ...



reviews.csv

TRANSFORM phase – Products



Dishwasher filter

Filtering of all dishwasher products: e.g. removing all entries with */accessori/* in the URL.



Name cleaning

Removing spurious sub-strings in the name, e.g. those containing dishwasher model, already stored in a dedicated field.



Claim detection

22 regex pattern to recognize claims based on *claim_raw* string. Creation of a boolean variable *has_** for each claim.



Output: stg_products.csv

csv file with the following columns: *code*, *name*, *series*, *install_type*, *energy_class*, *avg_rating*, *review_count*, *is_discontinued*, *url*, *claim_raw*, *has_**, *has_*_evidence*.



TRANSFORM phase – Products



Dishwasher filter

Removing reviews related to model codes not in *stg_products* (e.g. accessories).



Language and country

Split on *locale* field (e.g. *it_IT*), useful to extract a geographical dimension from reviews.



Numerical mapping

- *age_midpoint*: 'From 25 to 34' → 29
- *Household_size*: 'Couple' → 2
- *usage_weekly*: 'Seldom' → 0.1



Date parsing

From UTC format to *year, month, day*.



Review translation

Call to Google Translator through *deep-translator* library to unify reviews.



Output: *stg_reviews*

csv file with the following columns (selection):
queried_product, *date*,
rating_{overall, programs, loading, noise}, *text*, *country*,
age_midpoint, *household_size*,
usage_weekly.

LOAD phase – Data Warehouse



Relational DB

Electrolux standard (e.g. with respect to document DB, like Mongo), easily querable also by people with fewer experience.

Local implementation on SQLite for the Proof of Concept.



Variable selection

Only the most informative variables are stored in the the DB for the following analysis and visualization stages.

LOAD phase – Data Warehouse



PRODUCT_DB

[PK]	product_id	Bosch model code
	serie	Series 2/4/6/8
	install_type	Built-in / Freestanding
	energy_class	A – G
	avg_rating	Aggregated star rating
	review_count	total review number
	is_discontinued	bool
	has_aquastop, ...	22 claim flag (bool)

REVIEW_DB

[PK]	review_key	new unique primary key
[FK]	product_it	link to PRODUCT_DB
	year/month/day	date fields
	rating	1 – 5
	rating_progs, ...	4 secondary ratings
	country	2 letter ISO standard
	age_midpoint	15 – 68
	household_size	1 – 4
	usage_weekly	0.1 – 7
	title_en/text_en	review english translation

Data Warehouse – some numbers



PRODUCT_DB

- 141 dishwasher models:
 - 98 active models
 - 43 discontinued models.
- Good mix of built-in and freestanding models.
- Most represented energy classes: A and B.
- Most represented serie: Serie 4.

REVIEW_DB

- 15271 total reviews.
- 84 dishwasher model with at least one review.
- Average star rating: 4.84/5.
- Time span: from July 2017 to June 2026.
- 4230 incentivized reviews.



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Visualization tool disclaimer



- Electrolux doesn't have a visualization software (e.g. Microsoft PowerBI or Tableau) common to all employees.
- For similar tasks, pdf reports are usually employed.
- Therefore, the maps and plots shown in the following slides are generated from python code and embedded into the slides.

Review number by country

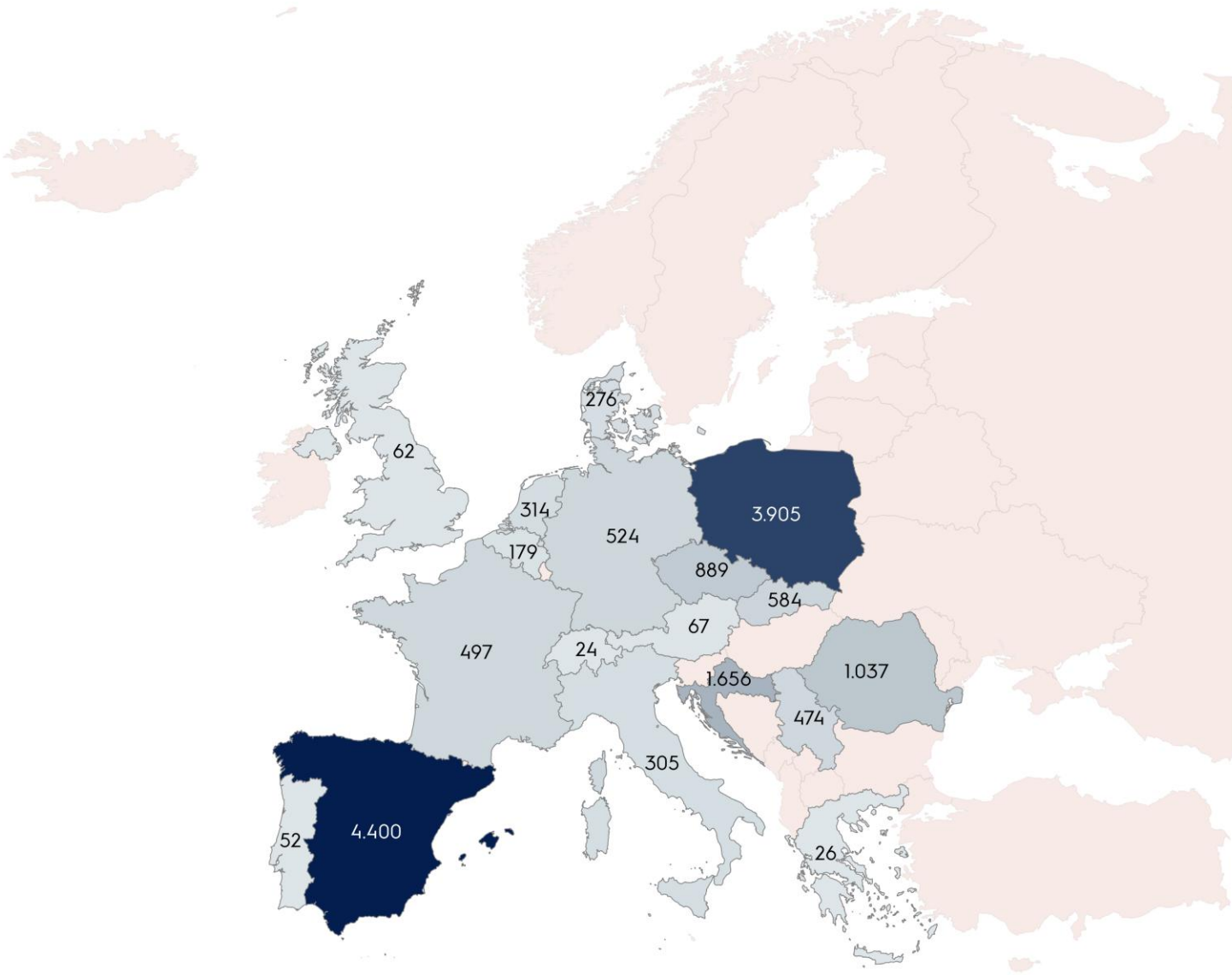


18 different countries are represented in the Bosch dishwasher reviews.

The top 5 countries contribute to almost half of the available reviews.

Country	Reviews	%
Spain	4400	17.4
Poland	3905	15.5
Croatia	1656	6.6
Romania	1037	4.1
Czech Republic	889	3.5

Bosch dishwasher reviews by country

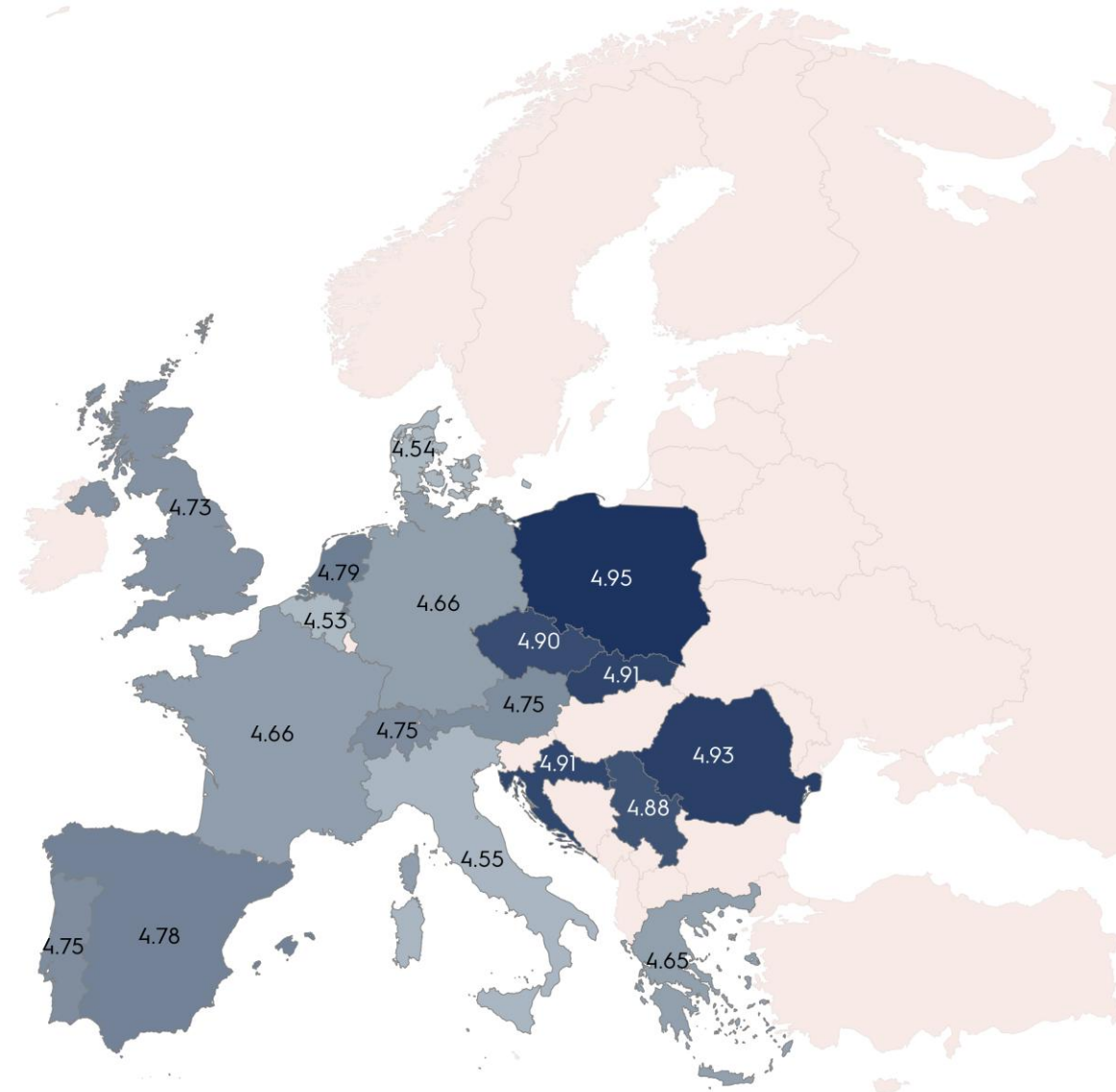


Star rating by country



Bosch dishwasher ratings by country

- On average, star rating is everywhere above 4.5.
- Countries from East Europe (Poland, Romania, Croatia, Slovakia, Czech Republic, and Serbia) tend to provide higher rates.
- Italy is one of the most severe countries.



Star rating evolution



With very few exceptions, star rating was constantly around 4.8 over years.

Incentivized reviews have lower variability, but don't seem to increase the average star rating. Hence, in the following incentivized and not incentivized reviews will be grouped together.

i

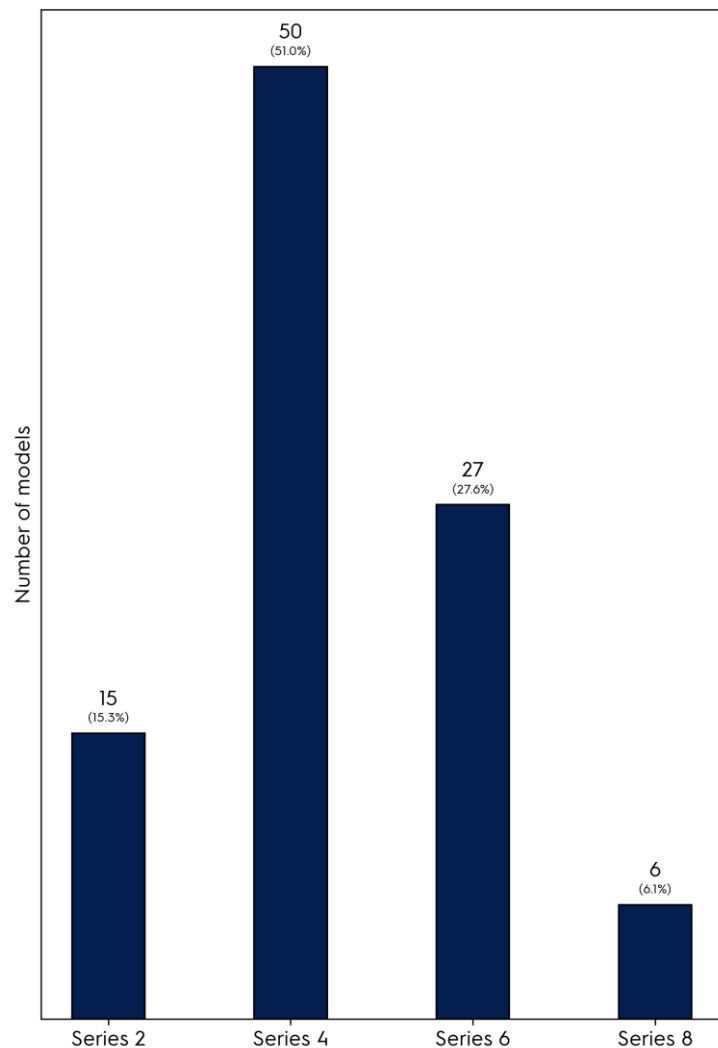
Axis limited in the range 4 to 5.

Every rating below 4 is usually considered as negative.

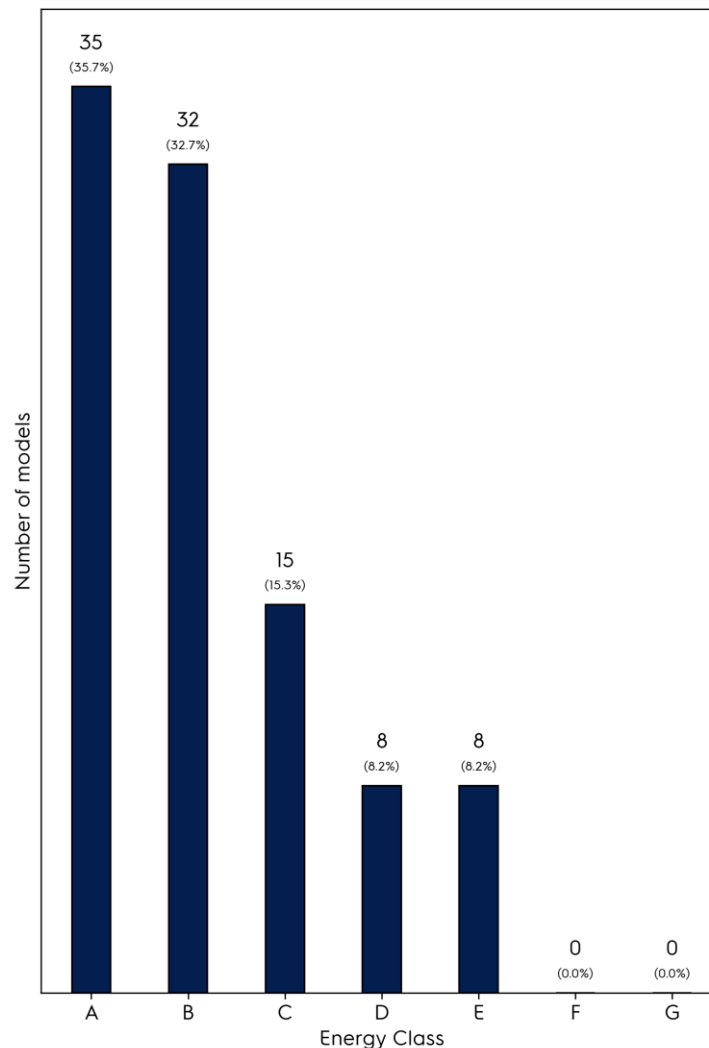
Bosch dishwasher average rating evolution



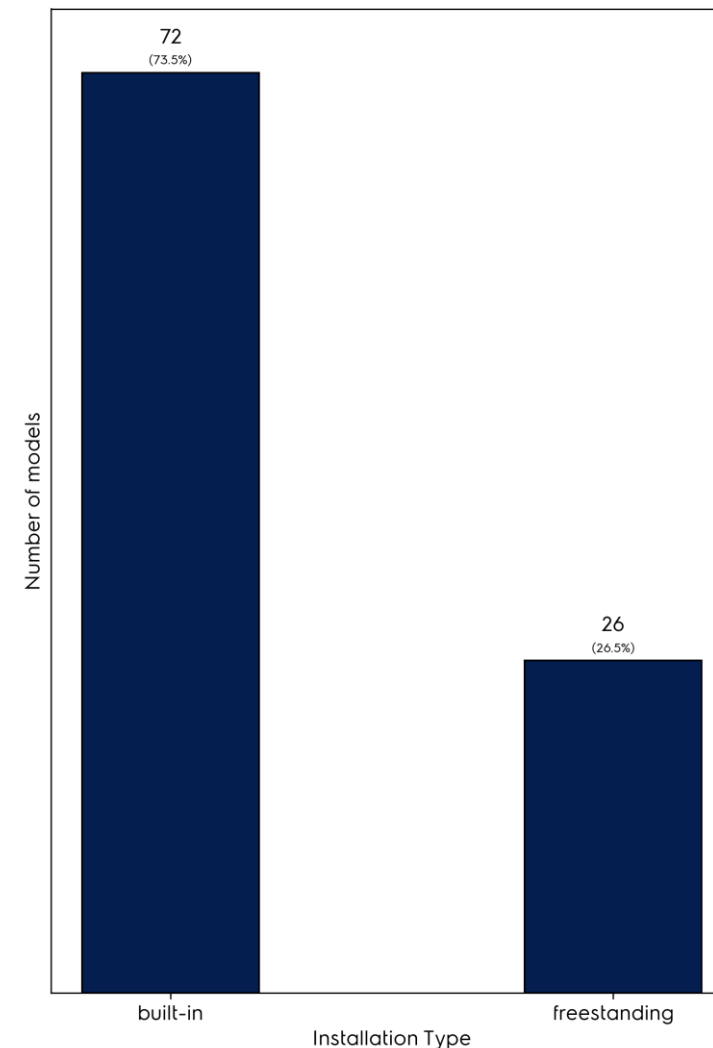
Bosch dishwasher roster

**Bosch dishwasher models by series**

Series 4: mass-premium appliances
Series 6: premium appliances

Bosch dishwasher models by energy class

A and B classes: market trends.
F and G discontinued by all manufacturers.

Bosch dishwasher models by installation type

Good mix between built-in and
freestanding appliances.

Bosch dishwasher roster



22 distinct claims on Bosch dishwashers:

Dosage assist

available on 100% of appliances, not a distinctive feature anymore.

Grazie al vano DosageAssist, il detersivo si scioglie in modo ottimale favorendone l'efficacia durante il lavaggio.



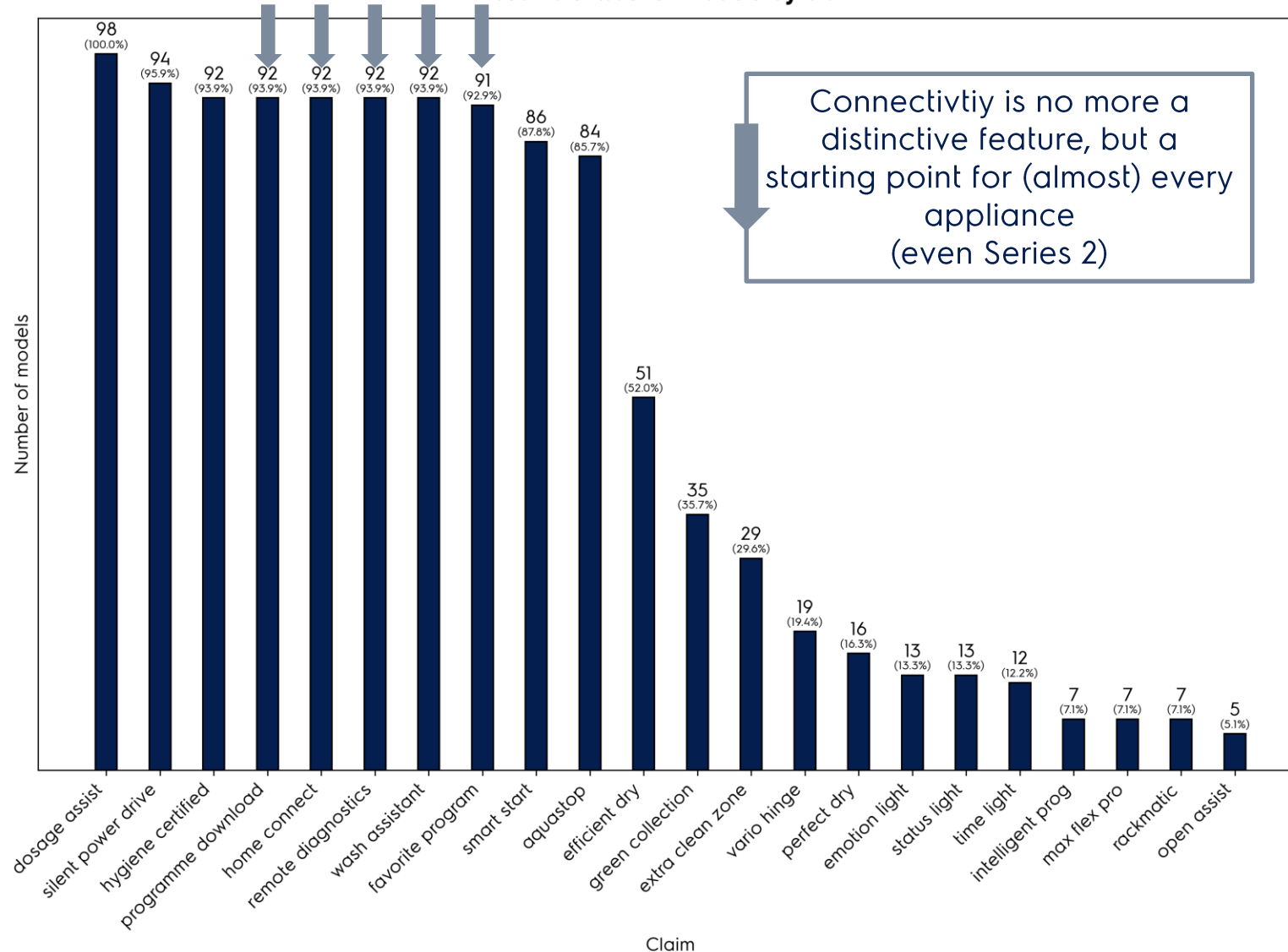
Open assist

The most exclusive feature, available on the most recent appliances only.

Open Assist: apertura facile, senza maniglia



Bosch dishwasher models by claim

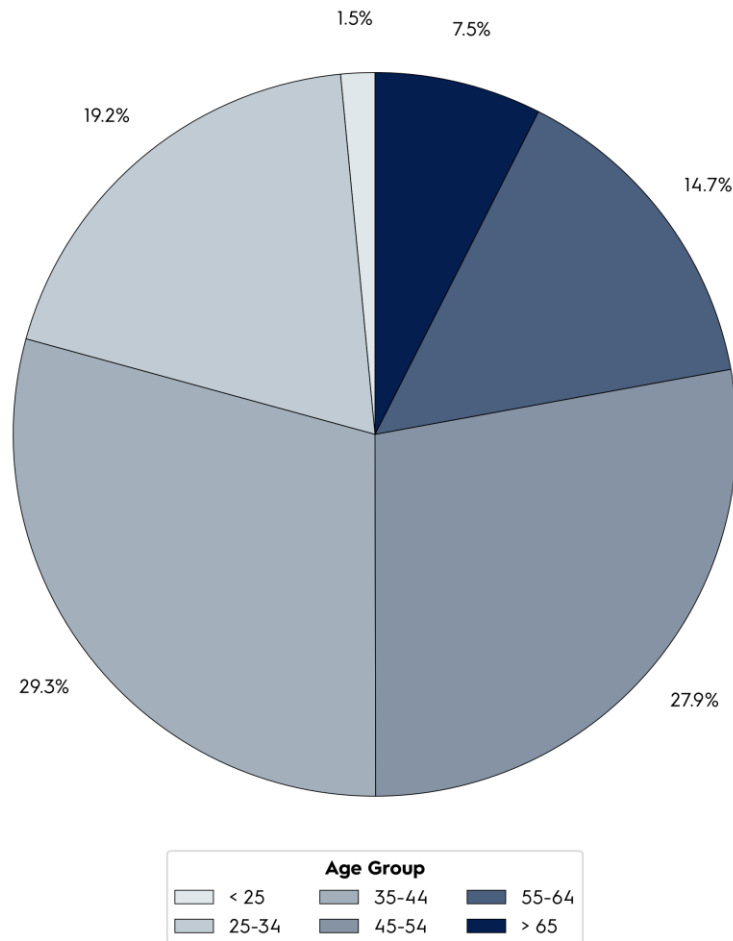


Connectivity is no more a distinctive feature, but a starting point for (almost) every appliance (even Series 2)

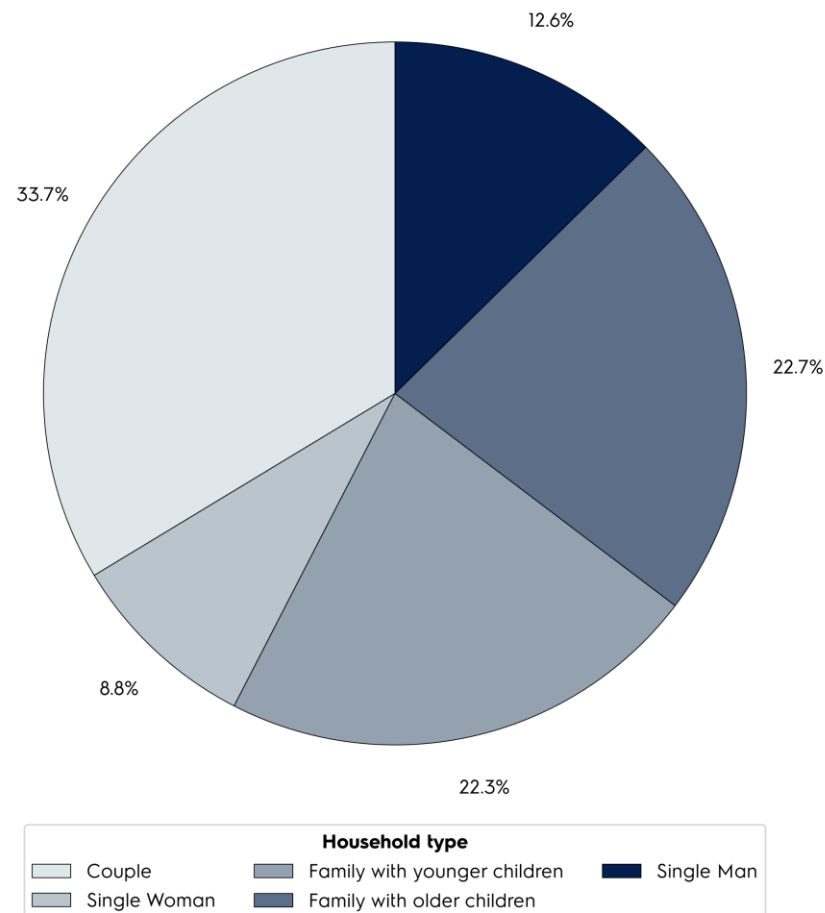


What does the Bosch customer look like?

Bosch dishwasher reviewers age distribution



Bosch dishwasher user household distribution

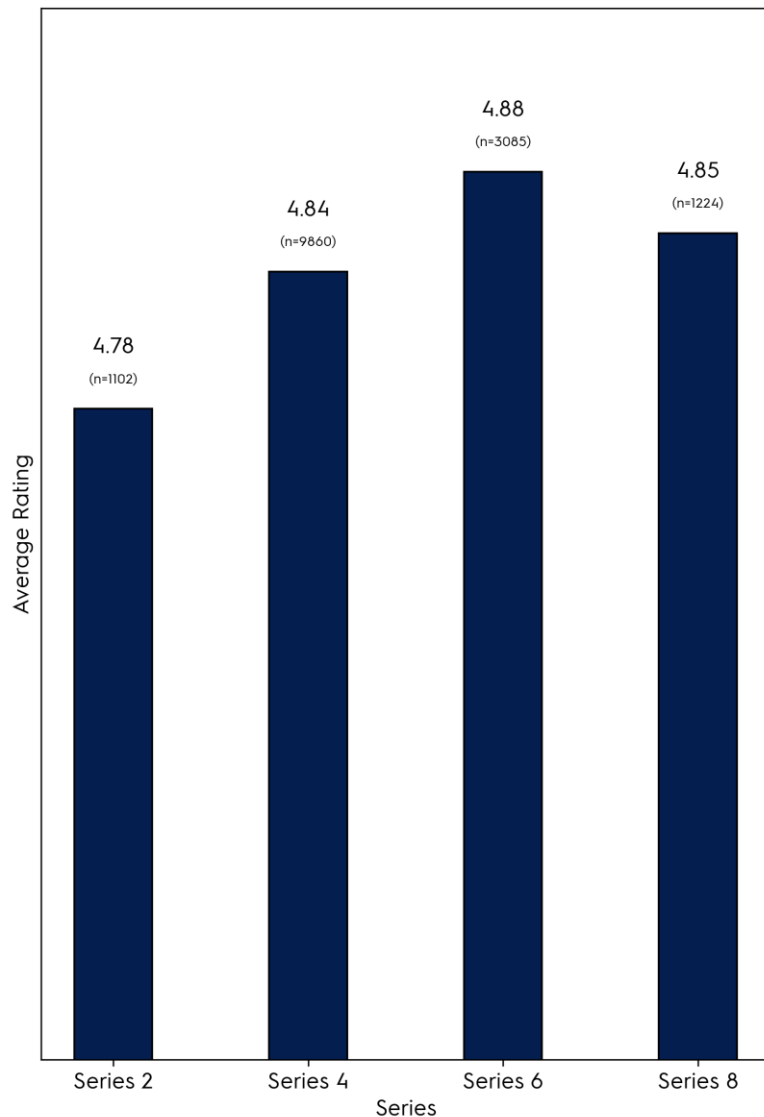


Bosch customer is usually in the age range 35 – 54 (57.2% of the data sample).
He is part of a family (78.6%), often with children (45%).
He uses the dishwasher almost everyday (67.9%).

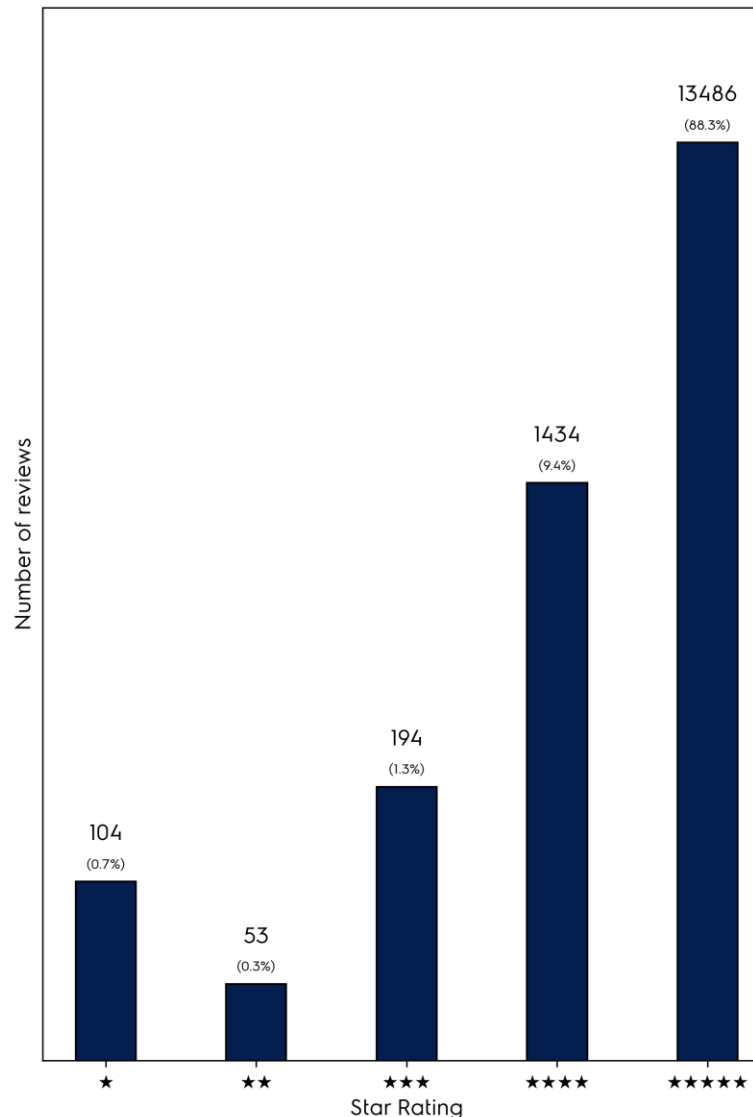


How are reviews distributed across series?

Bosch dishwasher ratings by series



Bosch dishwasher reviews by rating



97.7% of the reviews are *positive* (4 ★ or 5 ★), highlighting the high appreciation for Bosch products.

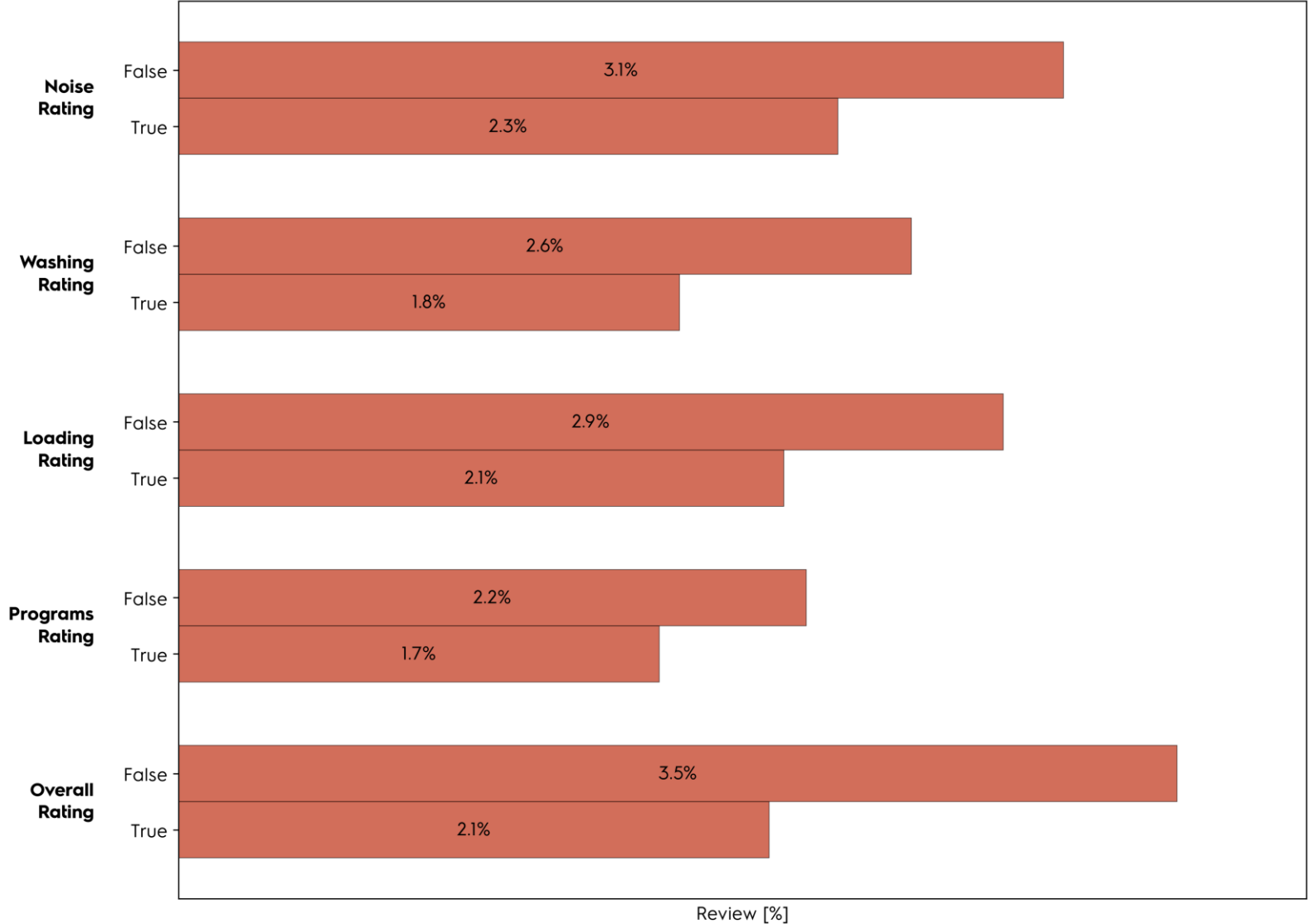
The lowest average rating is for Series 2 (4.78): it is the cheapest series, so probably its features are not perceived as fully aligned with the Bosch brand.

The highest average rating is for Series 6 (4.88): these appliances feature the best characteristics, without adding the Series 8 extra cost.



Claim effect on negative reviews: positive impact

Smart Start claim effect on negative (< 4) reviews



Some claims have a positive impact, helping reducing negative (< 4) reviews. Top 5 improvements are:

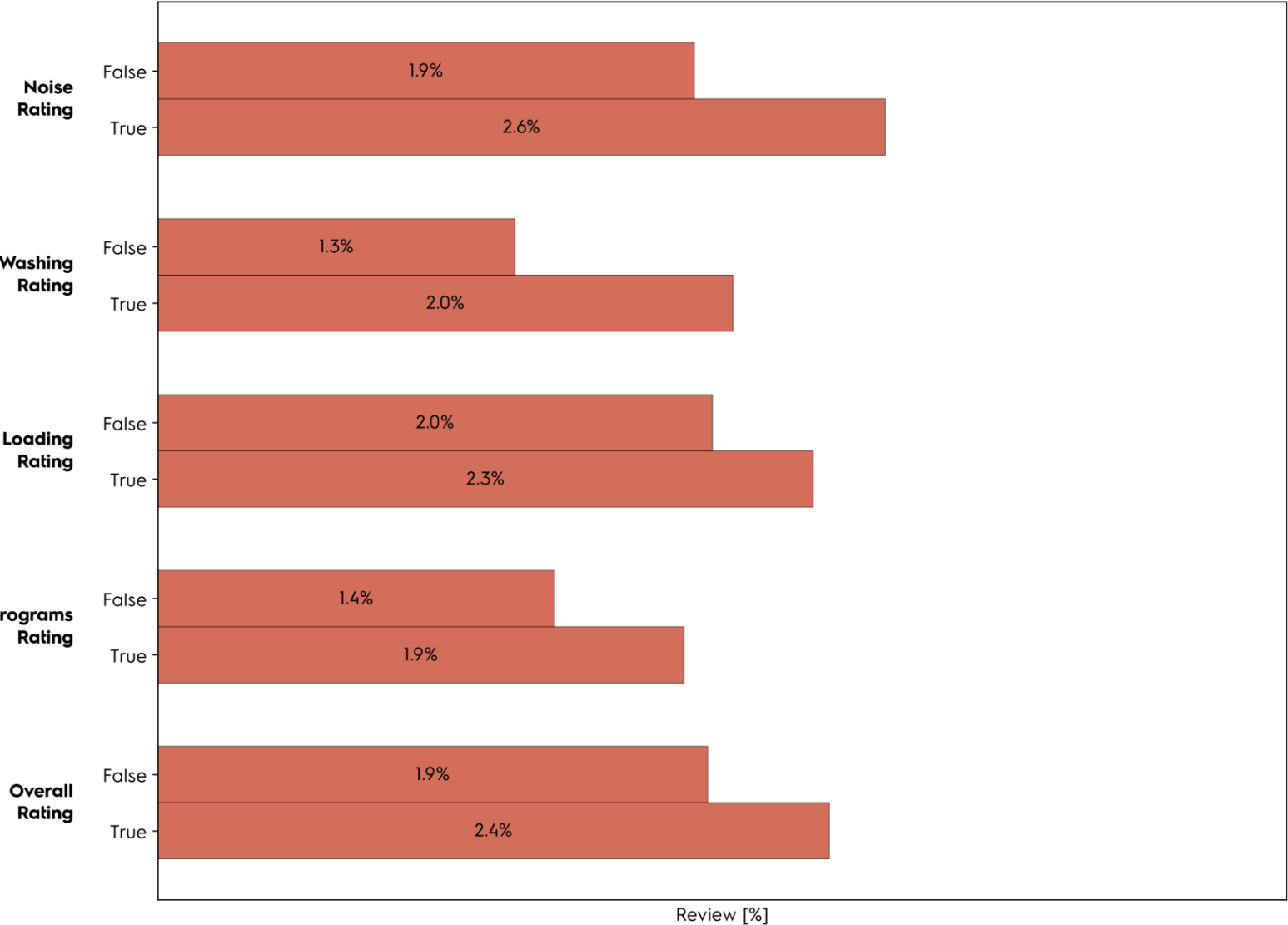
Claim	Rating	Δ reviews [pp]
Smart Start	Overall	1.45
Favorite Program	Overall	1.43
Program Download	Overall	1.36
Home Connect	Overall	1.36
Remote Diagnostics	Overall	1.36

The most impactful claims are all related to connectivity.



Claim effect on negative reviews: negative impact

Aquastop claim effect on negative (< 4) reviews



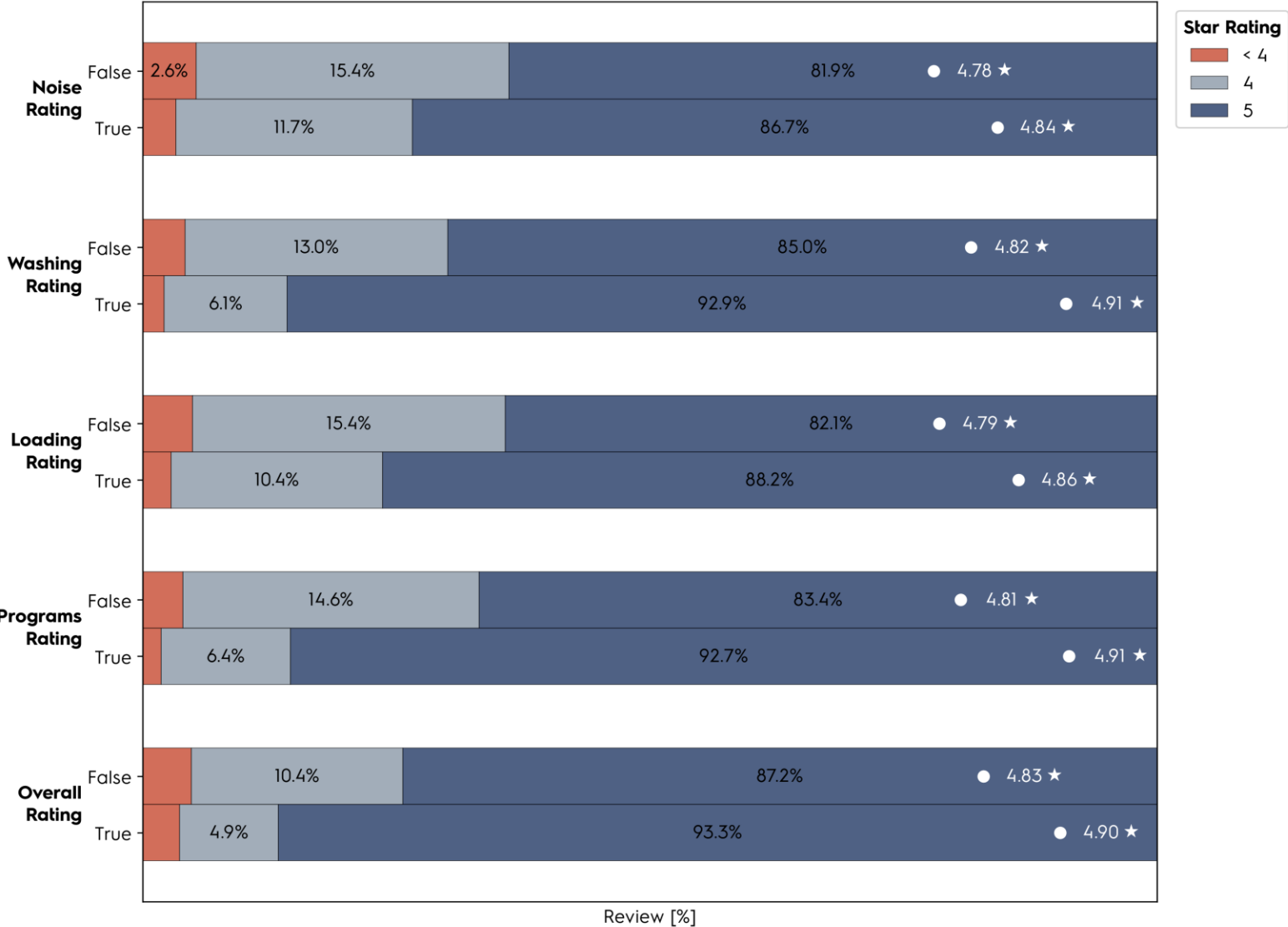
Some other claims impact negatively on the reviews. Top 5 negative impacts are:

Claim	Rating	Δ reviews [pp]
Aquastop	Washing	-0.77
Status Light	Overall	-0.70
Aquastop	Noise	-0.68
Open Assist	Noise	-0.50
Vario Hinge	Noise	-0.49



Claim effect on star rating: positive impact

Time Light claim effect on Bosch dishwasher reviews



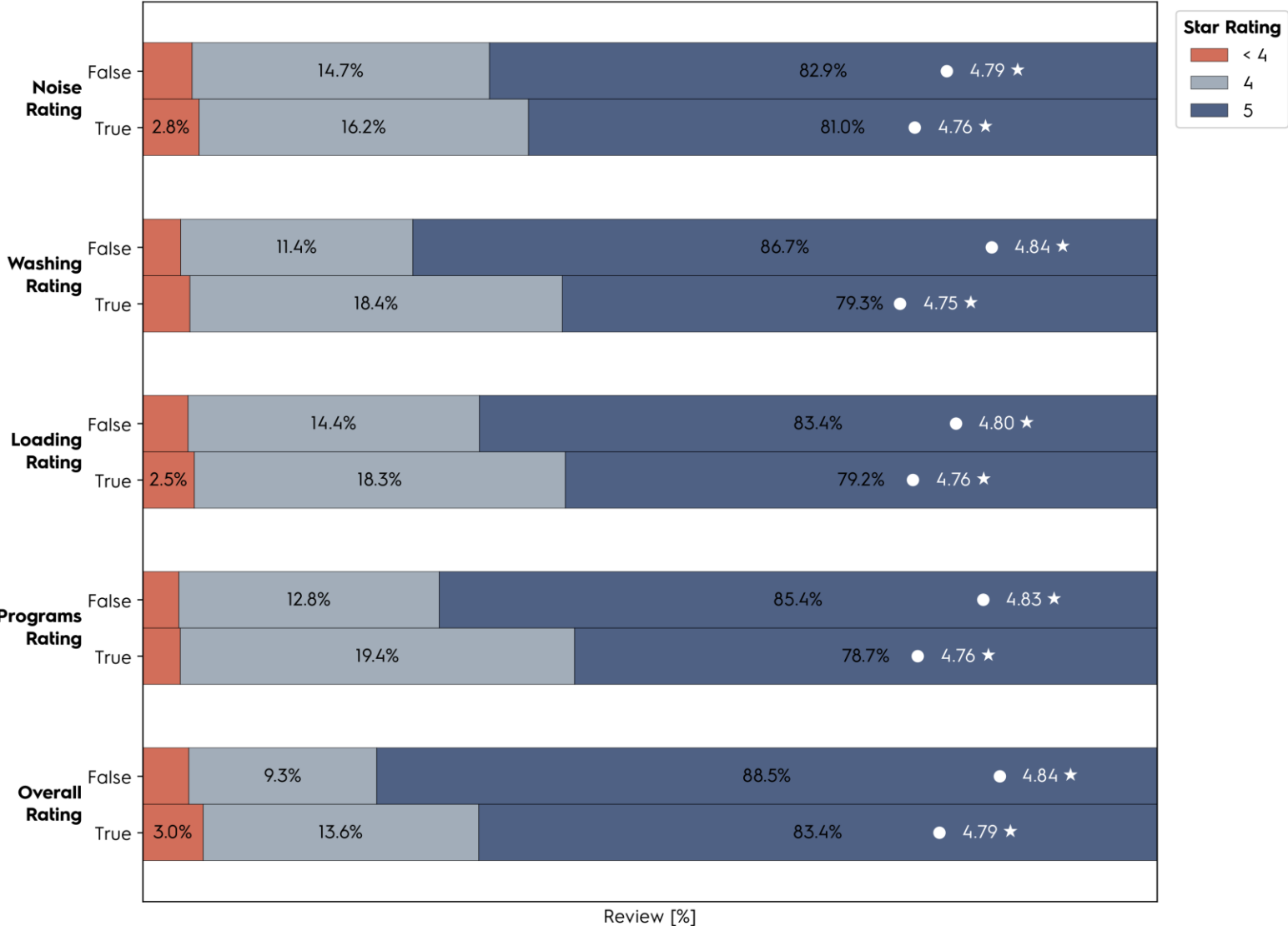
Washing and programs ratings are the most positively impacted by claims (also without any direct relationship, probably because some distinctive feature hardly fit into the available rating categories):

Claim	Rating	Δ rating
Time Light	Programs	0.107
Extra Clean Zone	Programs	0.095
Time Light	Washing	0.094
Emotion Light	Programs	0.091
Emotion Light	Washing	0.087



Claim effect on star rating: negative impact

Status Light claim effect on Bosch dishwasher reviews



Some other claims impact negatively on the rating. Top 5 negative impacts are:

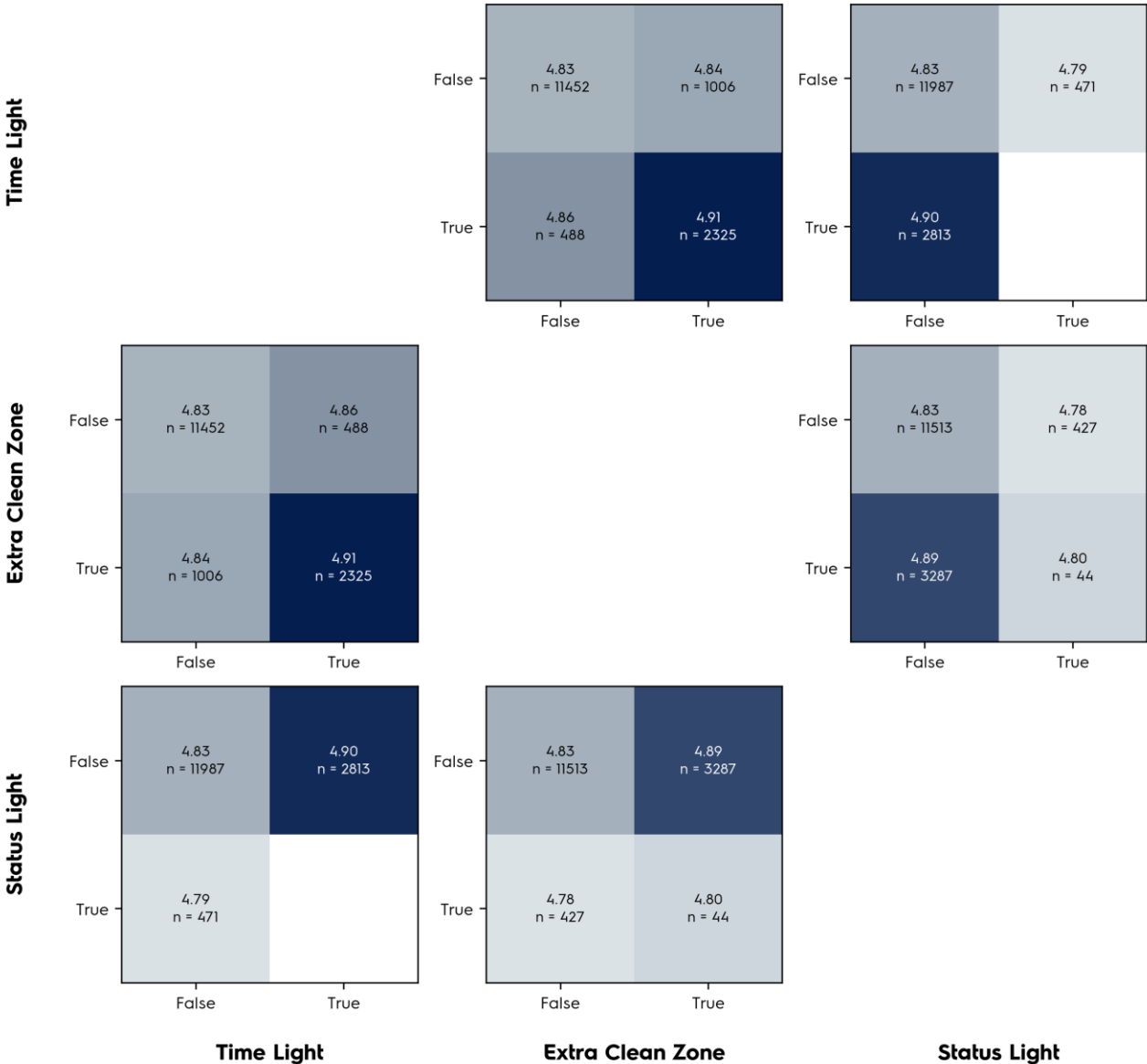
Claim	Rating	Δ rating
Status Light	Washing	-0.091
Vario Hinge	Washing	-0.078
Vario Hinge	Programs	-0.072
Aquastop	Washing	-0.069
Silent Power Drive	Washing	-0.068

Claim effect on star rating: joint effect



So far, the interaction between reviews and claims has been analyzed individually. However, even when examining the joint effect, the same conclusions can be drawn.

Claim interaction on overall rating





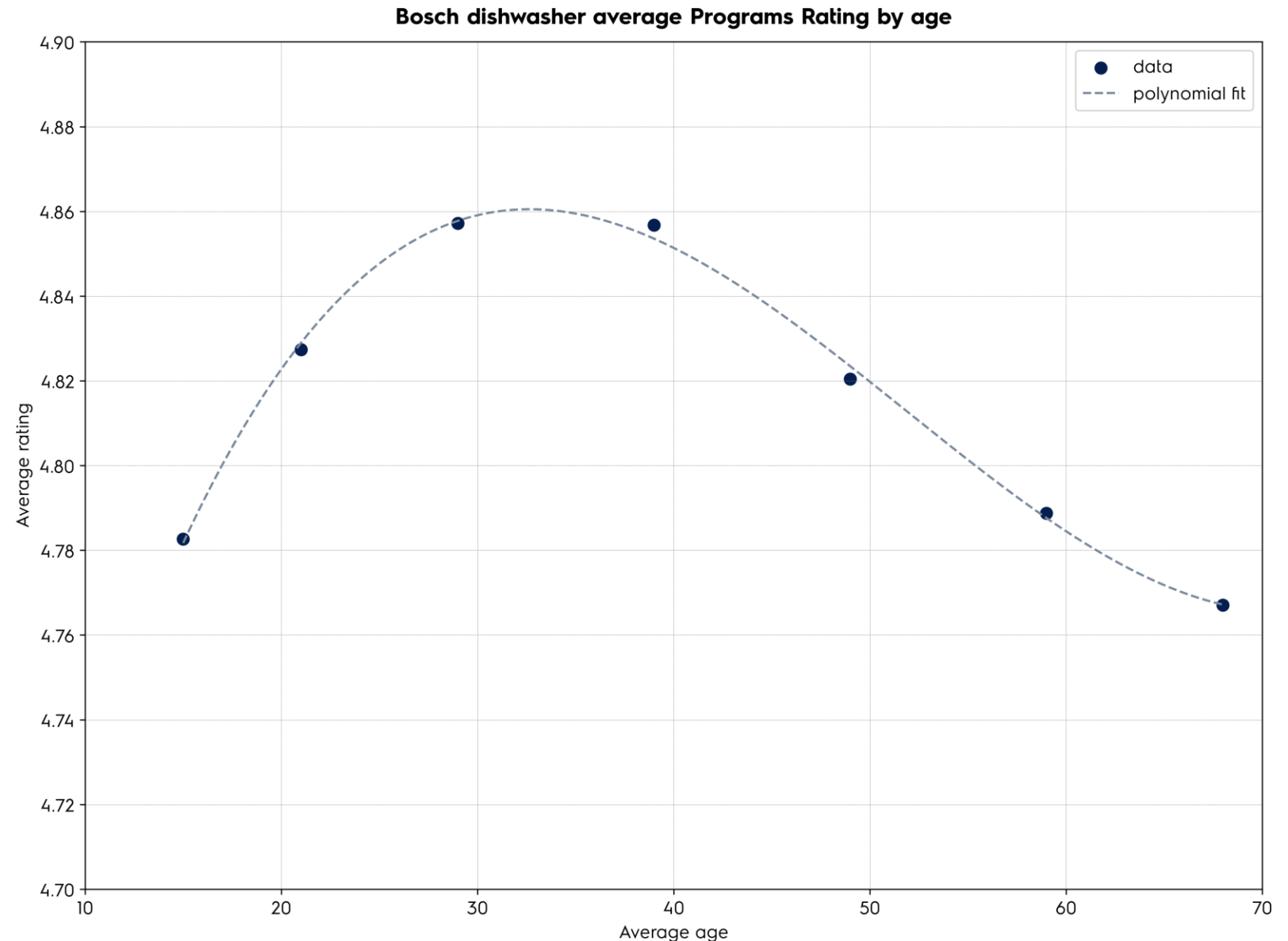
Claim effect on star rating: joint effect

Bosch customer characteristics influence the ratings.

Age is negatively correlated with all the ratings (the older the customer, the lower the rating).

Dishwasher usage is positively correlated with all the ratings: customers who use the dishwasher more, tend to give higher star ratings.

Family size is positively correlated with the ratings: the dishwasher is perceived as a real helper.





Are reviews and claims related?

Approach

1

Regex

Dictionary of regex for each claim, to understand if the reviews cite the exact claim name or some related topics.

2

Data Warehouse

Information on claims and reviews retrieved from the Data Warehouse previously built. Reviews for rating ≥ 4 only.

3

Review and claim correlation

Percentage of reviews citing the exact claim name or some related topic.

Claim	Reviews	Topics related to the claim [%]	Mention the exact claim [%]
Silent Power Drive	14146	39.3	0.0
Home Connect	13296	10.6	3.4
Open Assist	304	8.9	1.3
Efficient Dry	8782	6.9	1.0
Perfect Dry	3385	5.3	2.2

Hint



There is no direct correlation between claims and reviews. Customers leave comments related to claims, but they rarely mention them explicitly and often do so unconsciously. While claims may influence purchasing and ratings, they do not influence the content of reviews. Home Connect is mentioned more frequently because it is the name of Bosch app.



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Conclusions and next steps



Conclusions



ETL pipeline

The pipeline employed allowed to get insight on 100+ dishwasher models and 15k+ reviews from different countries, enabling a systematic analysis impossible from the raw scraped data.



Most impactful claims

Connectivity claims are almost ubiquitous in Bosch dishwasher and drive customer satisfaction, reducing negative reviews percentage.



Customer-profile dependency

Ratings are consistently above 4.5 ★, but scores are affected by customers age and habits, suggesting segmented marketing communication.

Next steps



Cloud migration

Employ cloud provider (e.g. AWS) to schedule the ETL pipeline and the visualization at regular intervals, e.g. biweekly or monthly. S3: storage, Glue: data integration and transformation, Lambda: services run in response to event.



Machine Learning analysis

Exploit review texts to perform sentiment analysis and classify reviews. Information available enables predictive models and clustering.



Extension to other competitors

Extend the approach to competitors in the same market segment (e.g. LG, Samsung, Beko, Whirlpool, ...)

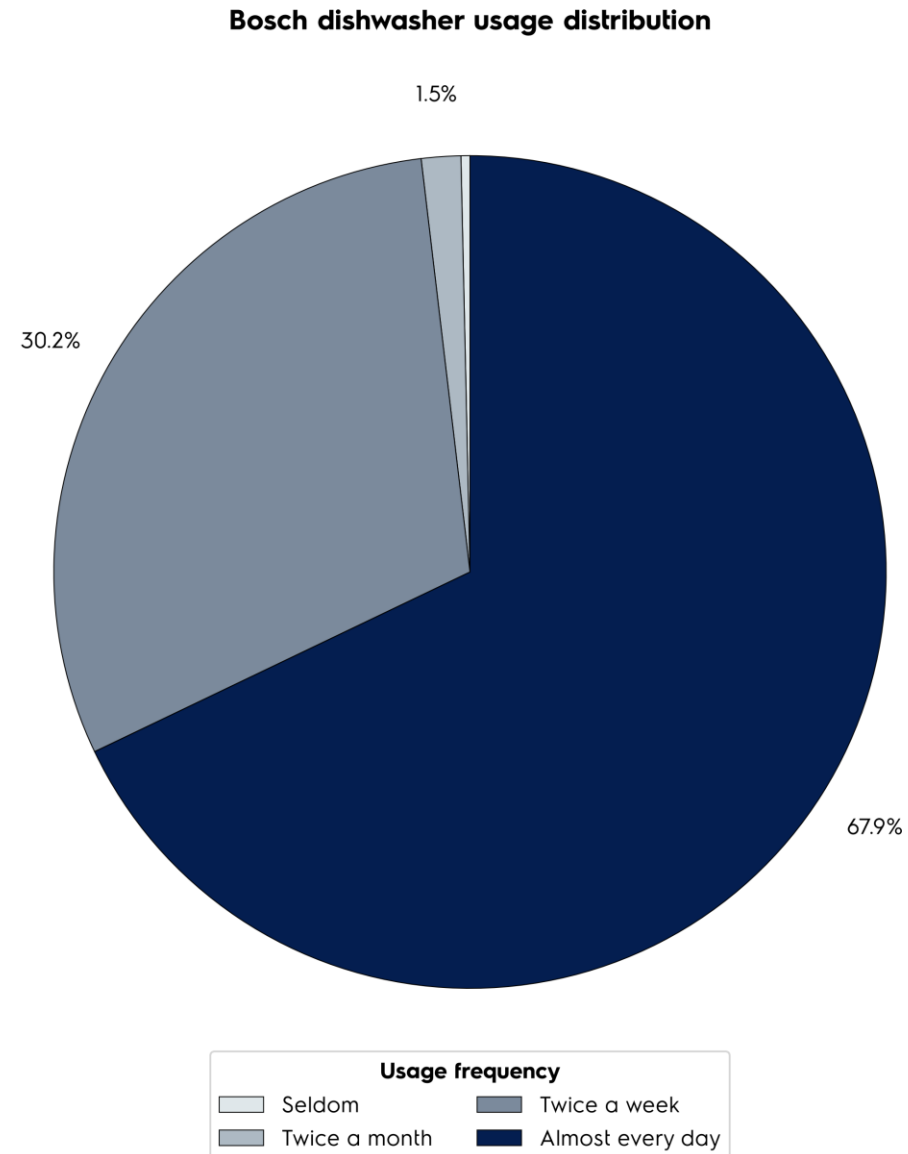


<https://github.com/SavarDar2204/Bosch-Reviews.git>



Electrolux
Group

What does the Bosch customer look like?



Bosch claims



Claim	Description
Dosage Assist	Thanks to the DosageAssist compartment, detergent dissolves optimally, enhancing its effectiveness during the wash cycle.
Silent Power Drive	SilentPowerDrive: powerful, quiet and long-lasting motor.
Hygiene Certified	Certified hygiene level for the ECO 50°C program, with even higher levels achieved with the Intensive 70°C and Machine Care programs.
Program Download	Program Download: additional wash programs for specific needs.
Home Connect	Home Connect: connected dishwashers for a simpler everyday life.
Remote Diagnostics	Remote diagnostics: in case of malfunction, call the service team and check the status of your appliance remotely.

Claim	Description
Wash Assistant	Wash Assistant: the right program for every load.
Favorite Program	Favorite Program: personalize your dishwasher.
Smart Start	Smart Start: strategic choice of the start time.
AquaStop	AquaStop: anti-flood protection system with Bosch guarantee against water damage – for the entire lifetime of the appliance.
Efficient Dry	Efficient Dry: thanks to the automatic door opening at the end of the cycle, the drying process is more effective.
Wash Assistant	Wash Assistant: the right programme for every load.

Bosch claims



Claim	Description
Green Collection	Green Collection dishwashers: use of low-CO ₂ -emission stainless steel. For a reduced environmental impact.
Extra Clean Zone	Extra Clean Zone: dedicated area in the upper rack for intensive cleaning of heavily soiled dishes.
Vario Hinge	Thanks to the Vario Hinge sliding hinges, the dishwasher can be installed using a plinth up to 5 cm in height, allowing greater flexibility in kitchen design.
Perfect Dry	PerfectDry: drying results up to 2 times better on plastic items.
Emotion Light	Emotion Light: interior lighting of the dishwasher.
Status Light	Status Light: the indicator light in the top left corner that shows whether the dishwasher is running.

Claim	Description
Time Light	Time Light: floor light indicator displaying the current wash phase and the remaining time until the end of the programme.
Intelligent Program	Intelligent Program: customize the wash program according to your preferences.
Max Flex Pro	Max Flex Pro racks: easy loading and protection for your dishes.
Rackmatic	RackMatic: upper rack adjustable in height across 3 levels.
Open assist	Open Assist: easy opening, handle-free.
Time Light	Time Light: floor light indicator displaying the current wash phase and the remaining time until the end of the programme.